

A Review of Greenhouse Gas Emissions from Agricultural Soil

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Abstract: Greenhouse gases (GHGs) like nitrous oxide (N₂O), carbon dioxide (CO₂), and methane (CH₄) are both emitted and removed by soils. Accurate worldwide allocations of carbon budget are essential for land use planning, global climate change, and climate-related research. Precise measurements, drivers, and mitigation strategies are necessary, given agricultural soil's significant potential storage and emission capacities. Different agricultural management practices cause greenhouse gas (GHG) emissions into the atmosphere and contribute to anthropogenic emissions. Agricultural soils can generate 70% of the world's manmade N₂O emissions and also behave as a CO₂ sink and a source of organic carbon and as producers and consumers of CH₄. When it comes to agronomic management, the source and sink of all these GHGs are distinct. Therefore, several approaches to measuring GHG emissions from agricultural soils are available and can be categorized into chamber systems and remote sensing approaches. Sustainable agriculture stands out as a viable and transformative approach to increase agricultural efficiency while addressing the challenge of GHG emissions. Incorporating advanced technologies, precise data analytics, and site-specific management practices can offer a pathway to mitigate GHG emissions, thereby reducing the global warming potential (GWP). Therefore, this review paper focuses solely on the drivers influencing and involving soil emissions and on quantification approaches for GHG emissions. In addition, mitigation practices aimed at optimizing GHG emissions from agricultural soils are highlighted.

Keywords: sustainable agriculture; soil emission; mitigation; agriculture



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1. Introduction

The amount of greenhouse gases (GHGs) in the atmosphere is extremely small in percentage but crucial to life on Earth. Without GHGs, Earth's surface long-wave radiation would instantly be back in space. In its place, the 161 W/m² radiation received and dumped into the environment by the surface of the Earth would be briefly stored in the atmosphere, bringing it to a livable temperature for life. Earth's surface would freeze at an average global temperature of −18 °C if GHGs were not there [1]. Thus, it is reasonable to assume that variations in greenhouse gas (GHG) concentrations in the atmosphere will result in changes in Earth's surface temperature, and radiative feedback will complicate the situation. The biosphere's temperature has risen in parallel with the rise in atmospheric GHG concentrations. Recent rises in global temperatures may be traced back almost entirely to rising levels of GHGs [2].

Soil serves a crucial function as both a contributor and an absorber of GHGs in nearly all land ecosystems. Strategies aimed at reducing emissions from soil are critical for determining optimal management practices in agriculture. GHG emissions, particularly nitrous oxide (N₂O), carbon dioxide (CO₂), and methane (CH₄) emissions, are increasing

worldwide because of anthropogenic activities in the agriculture sector [3]. These emissions play a significant role in modifying the environmental conditions by trapping heat from the lower atmosphere [4]. It is estimated that these factors account for the most significant part of modern climate change, with agricultural activities having a more significant impact on some GHGs than other factors, e.g., landfill and waste industrial processes and fugitive emissions [5]. Approximately 10–14% of all GHG emissions from humans worldwide are thought to come directly from agriculture each year [6]. CH₄ and N₂O are emitted in less significant amounts compared to CO₂, but their global warming capability is much higher, with CH₄ being 28 times more potent, and N₂O 310 times more potent [7]. Soil N₂O emissions account for 38% of agriculture's contribution to GHGs [8]. In contrast, manure N₂O emissions account for 7% of it, enteric CH₄ emissions from ruminant animals account for 32%, and N₂O and CH₄ from burning fossil fuels account for 12% [9].

Soil emissions make a substantial contribution to global warming potential (GWP) and play a crucial role in understanding and mitigating climate change impacts from agricultural practices. Soil emissions account for 21% of nitric oxide (NO), 53% of N₂O, 35% of CO₂, and 47% of CH₄, significantly contributing to GWP and underscoring the importance of accurately measuring these emissions for global budgets [2]. The production of CH₄ in the soil is contingent on anaerobic conditions. It is also influenced by factors such as carbon content and bulk density [10]. N₂O is generated through nitrification and denitrification processes, with transitions from wet to dry soil enhancing nitrification, and soil rewetting promoting denitrification, and has a GWP of 265 over 100 years [11]. The emissions from agricultural soils can be mitigated by adopting sustainable tillage practices, reducing soil disturbance, enhancing carbon sequestration, implementing cover cropping, and improving soil structure. Reducing the emissions of N₂O can subsequently contribute to a low GWP [12]. Sustainable agriculture techniques, such as site-specific nutrient management, optimizing fertilizer use, and integrating organic farming practices, can also contribute towards the mitigation of soil emissions [13]. These approaches not only lower GHG emissions but also improve soil health, increase crop yields, and contribute to sustainable agriculture development.

Agricultural soils can store significant organic carbon and function as both a source and a sink of GHGs [14]. Approximately 10% of atmospheric CO₂ passes through soils annually, underscoring the soil's role in GHG emissions [15]. Agriculture is a major contributor to GHG emissions, accounting for over 80% of N₂O emissions and 70% of ammonia (NH₃) emissions from human activities [16]. Furthermore, about 40% of anthropogenic CH₄ emissions result from enteric fermentation in agricultural activities [17]. Indeed, the soil is a pivotal component in the global carbon cycle and plays a critical role in the exchange of GHGs with the atmosphere. Therefore, the purpose of this study is to present a detailed account of how agricultural soils have added to the atmospheric burden of GHGs and of the key drivers of GHG emissions and, just as crucially, to demonstrate how sustainable management practices in the future can help reduce GHG accumulation in the atmosphere.

2. Significant Drivers of GHG Emissions from Agricultural Soil

GHGs are emitted in soils by microorganisms, root respiration, chemical breakdown activities, and the respiration of soil [18]. There are two types of drivers related to GHG emissions. The first category includes proximal factors that impact soil emissions within the immediate surroundings, such as local climate and soil type. The second category consists of distal drivers that influence soil emissions on broader scales, including factors like temperature and humidity [19]. These drivers are important in determining the GWP of agricultural soil. The critical drivers of GHG emissions from agricultural soils include land use/land cover change, nutrient availability, humidity, temperature, the soil potential of hydrogen (pH), fertilizers, and organic amendments. This study discusses these key drivers affecting soil GHG emissions.

2.1. Land Use/Land Cover

As per the Food and Agriculture Organization (FAO), approximately 31.5% of the global land area comprises grassland and pastureland, encompassing shrubs and small vegetation. Tree-covered areas account for 27.7%, while barren land, including semiarid regions, constitutes 15.2%. Cropland makes up 12.6% of the global land area, glaciers cover 9.7% of it, and water bodies occupy 2.7%; 0.6% of the global land area is categorized under various other classifications [20].

Vegetation cover is a significant factor in soil organic carbon buildup. Figure 1 presents the different land use/land cover of the global land area. Changing land use can have far-reaching effects on the source or sink attributes of CO₂ and other GHGs in the atmosphere and can impact the GWP [21]. Even though they only make up 2.7% of the land area, wetland areas have the highest average absolute emission rates. The lowest emission rates are seen in tree-covered areas, followed by grassland and pastureland, then croplands, including annual and perennial crops, and finally, bare regions [22].

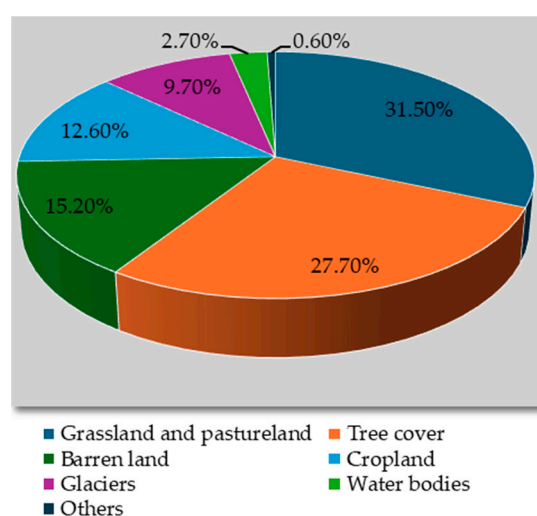


Figure 1. Total global land area with different land use/land cover [20].

Transforming forested, grassy, and peat lands into agricultural land substantially affects the emissions from soil related to changes in land use/land cover [23]. Figure 2 illustrates various categories of land use/land cover. The carbon content in the soil located in the top 7 cm experiences depletion within 30 years after converting forest land to agricultural land. However, no changes in the carbon content of soil are recognizable below plow depth. The total annual soil respiration can be increased dramatically from 24 to 57% in the first year following fertilization in semi-arid climates [24].



Figure 2. Land use/land cover categories: (a) cropland, (b) shrubland, (c) tree-covered area, (d) bare land, (e) grassland, and (f) peatland.

N₂O emissions from soil are affected by the types of wildlife grazing on pasture. There was a significant decrease in N₂O emissions in fields grazed by sheep compared to

those grazed by other livestock, while the lowest emissions were observed in non-grazed meadows. The authors of [25] analyzed N_2O and CH_4 emissions from seasonally dry ecosystems, finding usually low emissions of nitrogen oxides (NO_x) and uptake of CH_4 . Tree-covered and forested areas cover 28% of the total Earth's surface. However, vegetation cover and corresponding leaf size or area can differ greatly from grasslands with sparse vegetation to tropical rainforests with exceptionally dense foliage. In general, forests act as CH_4 absorbers. This is the most important finding of studies of CH_4 transport in many habitats [26]. During years characterized by low soil temperatures and notable GHG emissions, the soil respiration values tend to peak [27]. Any changes in land use/land cover can change the accumulation potential of soil organic carbon and can alter GHG emissions. Different types of land use/land cover such as grasslands, wooded shrub and tree landscapes, and desert environments have different rates of soil respiration in dry conditions. Compared to other land cover types, peatland areas have the highest total emission rates. The lowest emission rates are seen in tree-covered areas, followed by grassy ones, then croplands, and finally, bare regions.

2.2. Nutrient Availability

The correct respiratory activities of microorganisms and plants depend critically on the availability of nutrients. Therefore, the role of manure or fertilizer applications and of naturally occurring nitrogen and carbon in soil is significant [28]. If carbon is not a constraining factor, an elevated soil nitrogen concentration usually leads to a rise in net ecosystem exchange (NEE) [29]. Soil respiration becomes more sensitive to changes in the soil under these circumstances. Soil moisture increases after nitrogen fertilizer application and is less sensitive to changes in soil temperature [30]. In aerobic soil conditions, N_2O emissions were more significant after applying urea, but in saturated soil conditions, N_2O emissions were more marked after applying ammonium fertilizers [31]. Figure 3 illustrates the nitrogen losses from agricultural soil. Because not all nitrogen is usable by plants, reducing nitrogen emissions from agricultural fields requires optimized fertilizer rates according to the plants' needs. N_2O emissions rise with the excessive and unbalanced application of fertilizers [32]. The soil's water content also affects the selection of the right kind of fertilizer to prevent increasing N_2O emissions, which is also dependent on the tillage system.

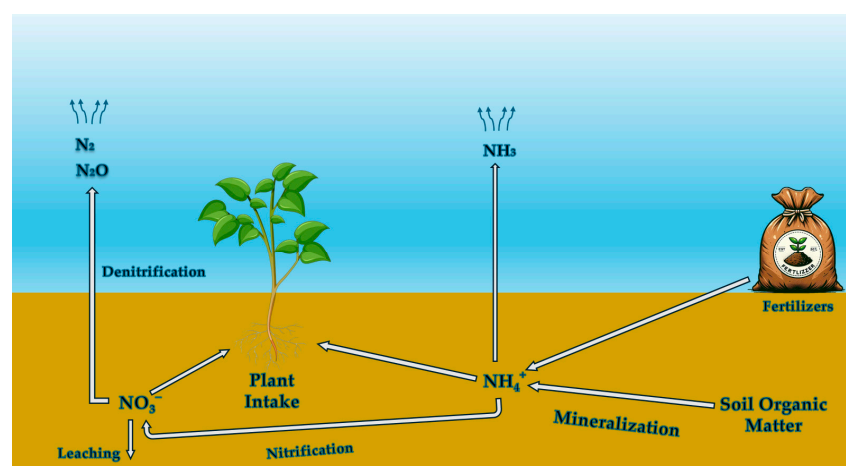


Figure 3. Nitrogen loss in agricultural soil.

When a farm switches from conservation tillage to the more sustainable no-till method, N_2O emissions fluctuate widely. Lowered soil emissions directly result from farm machinery usage, which causes soil compaction [33]. Soil nitrogen peroxide emissions in forests decreased as nitrogen deposition was reduced, while N_2O emissions remained unaffected.

2.3. Humidity

As it governs microbial activity and related processes, soil moisture stands out as the most essential soil attribute influencing soil gas emissions. Bacteria that convert nitrogen into nitrogen gas need oxygen, found in soil pores. Emissions from nitrification are most significant in soils with less than 20% of pore space filled with water [34]. A lower nutrient availability is associated with lower nitric oxide emissions in soils with less than 10% WFPS [35]. The capacity of nitrification to generate NO_x is greater than that to generate N_2O [36]. Strictly anaerobic conditions are necessary for CH_4 generation, which is positively correlated with soil humidity [37]. Soil emissions can be drastically decreased during extended drought. As a result, soils may become a net sink for NO_x [38]. Variations are brought on by rain that falls after a long drought. Within a few hours after precipitation begins, emissions rise and then, a few days later, decrease to the original levels [39].

2.4. Temperature

Variations in minor gas emissions from soils can be explained by the soil temperature, which is a key driver of GHG emissions. N_2O emissions can be increased by 74% due to soil moisture and by 86% due to temperature [40]. Higher soil temperatures also result in more significant emissions and higher soil respiration rates as a reaction to enhanced microbial activity. Increasing soil temperatures cause increased respiration rates, reducing soil oxygen concentration and further increasing the pressure for CH_4 and N_2O release [41]. Since water is necessary as a passageway for essential micronutrients, soil water stress may act as a layer above the beneficial effect of warmth [30]. Emissions of N_2O and CO_2 rise exponentially with rising temperatures. In the field, it can be tricky to see distinct relationships between the effects of temperature and those of moisture [42]. An estimated 50% of annual N_2O emissions can be affected by temperature changes related to freeze–thaw episodes that force gas release through agricultural soils [43].

2.5. Soil pH

Soil acidity can inhibit microbial activity. Liming is an example of a management strategy that affects soil GHG emissions by causing an increase in carbonate released as CO_2 [44]. When the soil pH is low, fewer emissions are produced. CH_4 generation is most effective at pH levels between 4 and 7 [26]. At neutral pH, CO_2 emissions were found to be at their greatest [45]. N_2O emissions can be affected by soil pH. The balance between NH_3 and nitrates (NO_3^-) shifts, producing ammonium cation (NH_4^+), at higher pH values, and nitrification increases as the pH rises [46]. Denitrification generates NO emissions in acidic soil. Nonetheless, the pH value was not shown to be significantly correlated with either nitrogen peroxide or N_2O emissions [28].

2.6. Fertilizers and Other Amendments

Between 1970 and 2010, the consumption of synthetic fertilizers rose by 200–300%, mirroring the expansion in food production [28]. Projections suggest that from 2022 to 2026, the yearly requirement of nitrogen will grow by 1.2%, that of phosphorus (P_2O_5) by 1.8%, and that of potassium (K_2O) by 1.6% [47,48]. Despite the benefits provided by chemical fertilizers, their incorrect use can negatively impact freshwater and terrestrial ecosystems. It can lead to nutrient runoff, soil nutrient depletion, eutrophication, diminished biological diversity, acidification, and a significant increase in GHG emissions resulting from agricultural activities. The energy-intensive and GHG-emitting nature of fertilizer manufacturing is also noteworthy, as it currently accounts for approximately 1.2% of GHG emissions [49]. Figure 4 shows the region-wise breakdown of total nitrogen consumption around the globe. Certain sulphate fertilizers, oxidants, and soil dressings can influence microbial activity, which can lead to GHG emissions. Soil dressings like lime or gypsum can modify soil pH and structure, influencing GHG emissions indirectly [50].

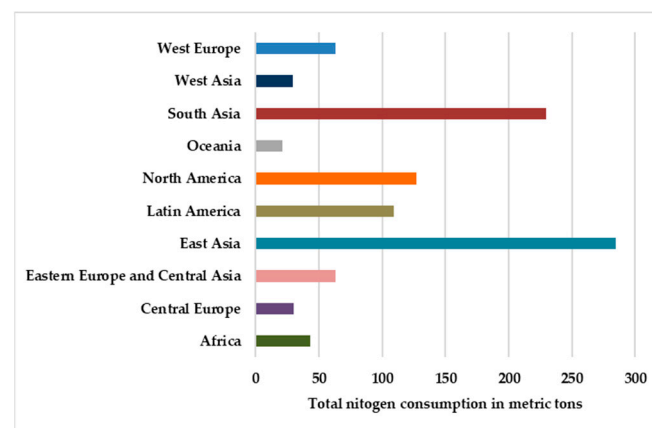


Figure 4. Region-wise breakdown of total nitrogen consumption in metric tons of nutrients [51].

2.7. Organic Amendments

The application of organic amendments can contribute to the release of GHGs, further increasing environmental concerns [52]. The organic matter decomposition process in soils causes the emission of gases such as CO_2 and N_2O , both potent GHGs that contribute to climate change. For example, animal and composting manure significantly affect GHG emissions. It was observed that poultry manure resulted in a noteworthy increase in CO_2 , CH_4 , and N_2O emissions compared to dung from pigs and cows [3]. The application of this manure led to a substantial rise in NO_2 emissions, with an average increase of 32.7% compared to that recorded when using a synthetic nitrogen fertilizer alone [53]. Employing organic agricultural practices has been identified as a means to enhance carbon sequestration [54]. The production of CH_4 and its release from animal manure result from a combination of anaerobic conditions and the presence of degradable organic matter in manure. It is worth noting that manure from livestock contributes approximately 6% of CH_4 emissions due to human activities. The efficacy of compost in agriculture faces limitations due to the loss of carbon and nitrogen in the atmosphere, restricting its agronomic effectiveness and contributing to greenhouse gas emissions [55,56]. In situations where composting processes are not managed appropriately, there is a risk of heightened GHG emissions [57]. A study conducted during the composting of cow manure revealed that nitrogen loss through ammonium (NH_4) volatilization ranged from 19 to 42% of the total nitrogen, while carbon loss as CO_2 varied from 46 to 62% of the total carbon [58]. NO_x , a byproduct of nitrification and denitrification processes mostly due to bacterial activity, is also generated during composting. Additionally, compost windrow facilities, as indicated in [59], have significant impacts, particularly on CH_4 and N_2O , which are released into the atmosphere. Therefore, while organic amendments offer benefits in terms of soil health and fertility, it is crucial to carefully manage their application to mitigate the potential drawbacks associated with nutrient eutrophication and GHG emissions.

3. Measurement of GHG Emissions

GHG emissions from soils involve direct observations in both field and laboratory settings. There is no universally superior technique to measure them, and the choice of the method depends on various factors such as application type (considering the observable area, analyzing procedures, continuous monitoring, and spatial variability), accuracy and precision of the technique (including bias and potential influence on soil structure), as well as associated costs and workload. Employing multiple methods is often advisable, considering the above-mentioned applications and considerations related to accuracy and precision when measuring GHG emissions.

3.1. Chamber Systems

The analysis of soil emissions, including CO₂, CH₄, N₂O, and NO, commonly employ flux chamber-based techniques [60–63]. This method involves placing a box or a cylinder on the soil surface, allowing its base to be open at the ground. Therefore, the gases released accumulate in the chamber headspace, and alterations in their mixing ratio are examined through the use of diverse gas sensors, including gas chromatography and infrared spectrometry. To address potential challenges related to uneven gas concentrations within the chamber, measures such as reducing chamber height and employing lower detection limits can be taken [64,65]. The results obtained with this method provide data on the spatial scale, representing measurements at small spot dimensions. Nevertheless, for a thorough assessment of site-level fluxes, the chamber system has to be swiftly and easily relocated to measure multiple pre-defined locations [66].

The chamber systems are classified into closed and open chambers (refer to Figure 5a,b), with the closed chambers further divided into closed dynamic chambers and closed static chambers [67]. The closed dynamic chambers, referred to as non-steady flow-through chambers, may lack standardization, impeding direct comparisons of datasets from various research groups [62]. All chamber systems have to integrate auxiliary sensors to capture parameters influencing the emissions from soil. Relative humidity, pressure, and air temperature sensors need to be set up inside and outside the chamber to record the ambient conditions and detect discrepancies within the chamber. To ensure precise NEE measurements, an outside photosynthetically active radiation (PAR) sensor should be incorporated [68]. Further, every chamber must be positioned on a collar (made of the cost-effective polyvinyl chloride or of steel) to prevent gas leakage. To minimize the collar's impact on soil structure and plant roots, it is advisable to embed it a few centimeters deep [60]. The soil profile is influenced by the collar, and consequently, the measurements of flux necessitate its installation at least 24 h before the initial measurement [69]. Although certain chambers may not need a collar, this approach is not advised for use in forests. To explore gas production, the profile of gas concentration can be examined at various soil depths [70]. Samples from certain soil depths can be evaluated in the laboratory using gas chromatography [71], or gas sensors can be installed at particular soil depths for continuous and automated measurements [72]. Closed static chambers are widely used for CH₄ and N₂O but are less accurate for CO₂. Manual systems are labor-intensive but cost-effective. The closed dynamic chambers are versatile, being able to measure multiple gases with options for continuous monitoring, but they have higher costs and operational complexity. In contrast, the open dynamic chambers can provide continuous, accurate measurements without accumulation times. They are also best suited for challenging environmental conditions but have higher costs and complexity.

An eddy covariance method directly approaches the micrometeorological process, which uses vertical turbulence between the soil surface and atmosphere to measure the interchange. The method scrutinizes the gas interchange, and a gas analyzer should be installed at least 2 m above the ground. The most frequently examined gases are CO₂, CH₄, and N₂O; other chemicals, such as carbonyl sulphides, can also be identified [73,74]. This approach fully accounts for the soil, biosphere, and atmosphere by integrating plants and trees to determine the NEE. The process functions well only in the presence of turbulent mixing occurring near the ground. This system can lead to flux underestimation, the post-processing of the data is complex, and gap-filling measurements are also critical for calculating the gas flux [75]. Eddy covariance systems frequently experience issues at night and during times of little disturbance. The open-path Fourier Transform Infrared Spectroscopy (FTIR) approach is applicable at night and when there is no turbulence. This technique is particularly apt in situations where the entire ecosystem, including larger plants and trees, cannot be accommodated within the confines of a chamber [76,77].

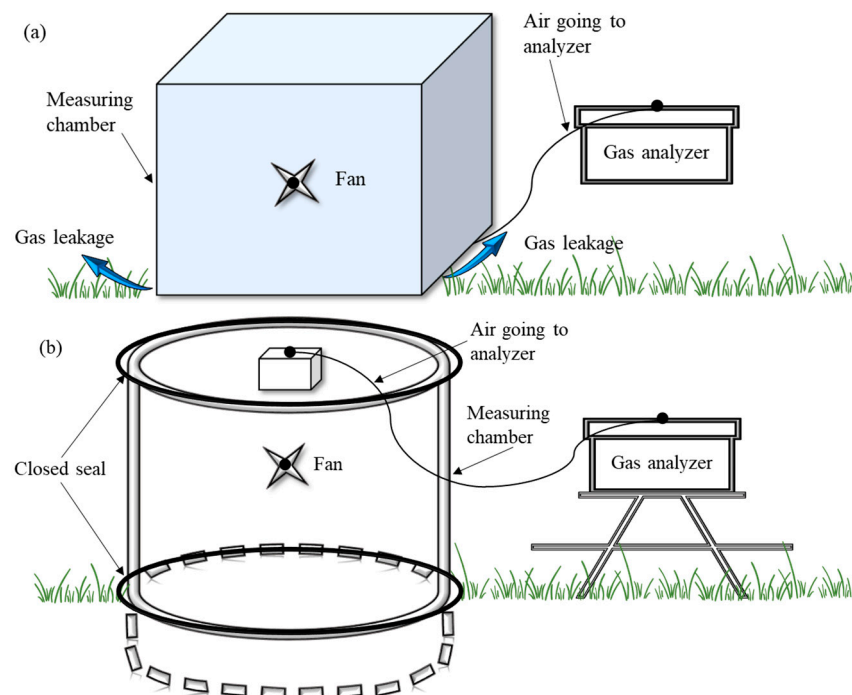


Figure 5. Illustration of (a) open chamber system and (b) closed chamber system.

3.2. Remote Sensing

Two remote sensing methods from satellites might offer data on GHG soil emissions. An approach entails gauging the intensity of sunlight reflection within specific narrow wavelength bands in the spectrum's visible and short-wavelength infrared segments. These data are subsequently used to ascertain the tropospheric and near-surface CO₂ and CH₄ concentrations. In the initial stages, European Space Agency (ESA) Earth observation missions like that employing the European Remote Sensing Satellite-1 (ERS-1) utilized imaging absorption spectrometers designed for atmospheric mapping, achieving a precision of 1–2% [78]. The Carbon-Monitoring Satellite aimed to extend the global time series of CO₂ and CH₄ concentration measurements. An anticipated mission for the EarthExplorer 8 satellite, scheduled for launch later in this decade, is CarbonSat [79]. The satellite's orbit or the plane's flight frequency and track will determine the collection of remote sensing and airborne data. However, the measuring network across oceans and the tropics will be sparse. Since it is still tricky to tell sources from sinks, additional remote sensing data are required [80]. Remote sensing methods become essential in situations where the measurement of entire ecosystem fluxes is impractical due to the limitations of the chamber systems, as large plants and trees cannot be accommodated within them. GHG emissions using drones is an innovative and efficient approach that allows for more accurate and timely data collection. Drones or unmanned aerial vehicles (UAVs) can be equipped with various sensors to capture data related to GHG concentrations. A study [81] was conducted on the use of high-resolution imagery from a drone to map nitrogen deposition hotspots in a dairy cow-grazed grassland. Combining these data with N₂O emission measurements, the study estimated the annual N₂O emissions and found them significantly lower than the default Intergovernmental Panel on Climate Change (IPCC) estimates. The method helps identify areas for targeted management to mitigate N₂O emissions and improve fertilizer application strategies. Different sensors can be mounted on these UAVs to measure GHG emissions [82].

3.3. Comparison between GHG Measurement Methods

We compared these methods based on their spatial viability and observational area, but no single technique stood out as the best. Investigations can incorporate combined

approaches, e.g., combining chamber systems, eddy covariance, and remote sensing, as per the output requirements. Table 1 presents a comparison between the different GHG measurement methods for agricultural soil.

Table 1. Comparison between GHG measurement methods.

GHG Estimation Method	Spatial Variability	Observable Area	Advantages	Challenges
Chamber systems	Ideal for emissions from distinct sites with a spatial footprint of less than 1 m ²	Can cover larger areas (up to 10,000 m ²), the number of chambers must be increased, or the chambers must be displaced consecutively	Useful for detailed studies of agricultural management practices on soil or plants	Require careful selection of representative sampling. High workload due to manual setup
Remote sensing	Large spatial extent	Can cover very large areas, on the regional and global scales	Extensive coverage. Frequent data collection	Sparse data in specific regions (e.g., oceans, tropics). Requires ground-based validation

4. Mitigation of GHG Emissions from Agricultural Soil

About 10–14% of all GHG emissions worldwide are attributed to agriculture [6]. N₂O, CH₄, and CO₂ are the primary GHGs produced by agricultural management methods [46]. The major task of agriculture is to enhance productivity without harming the environment, because sustainable food production concerns society [83,84]. Consequently, implementing low-carbon industrial models is crucial to reducing and eliminating GHGs from the atmosphere [85,86]. To reduce GHG emissions from agricultural soils, it is essential to decrease N₂O, CO₂, CH₄, and CH₄ emissions and improve carbon sequestration in soil. The decrease in N₂O emissions and rise in carbon sequestration in crop agriculture are the fundamental mechanisms to lower the GHG levels. Since soil carbon can be produced from the decomposition of soil organic matter, organic fertilizers also substantially impact the soil CO₂ and CH₄ emissions. Nevertheless, it was demonstrated that applying both organic and inorganic fertilizers can lower the soil GHG emissions. Crop management systems, such as using cover crops and different crop systems, intercropping, organic fertilizers, and tillage management, can reduce emissions and boost carbon sequestration.

4.1. Bio-Resource Management

Soil organic matter is extensively considered a critical component in assessing how well agricultural management strategies function in promoting sustainable agriculture, regardless of whether they are derived from biochar or from bio-digestate. The authors of [87] investigated the relationships between agricultural techniques and various environmental elements, such as plant elements. Their findings demonstrated that different levels of long-term management techniques have other effects on the environment and soil characteristics. A study [88] determined that the incorporation of organic amendments, such as sludge from sewage and municipal waste, agricultural residues, manure, and feces, can add to soil organic matter by as much as 90%. When an organic fertilizer was applied alone, soil respiration was encouraged; however, soil carbon storage was enabled when organic and inorganic fertilizers were used together, and GHG emissions were decreased. When organic and inorganic fertilizers were used together, carbon emissions were reduced compared to when the organic fertilizer was used alone [89]. Sustainable agriculture promotes environmentally friendly and low-cost farming techniques by using native microorganisms. This highlights the importance of collaborating with natural processes to conserve resources like soil while minimizing production costs and waste generation, which will lead to improving soil health, not only reducing GHG emissions but also helping to improve soil carbon sequestration [90].

4.2. Nutrient Management

The persistent issue of excessive, incorrect, and uneven application of fertilizers and nitrogen in agricultural fields has raised considerable concern. Improving the efficiency and usefulness of nitrogen utilization for crops has the potential to reduce N_2O emissions by mitigating the likelihood of increased residual NO_3^- in the soil [91]. This can be achieved by using the 4R nutrient management practice, which includes the right source, right time, right rate, and right placement [92]. Slow-release fertilizers and nitrification inhibitors can also play an important role in the reduction of GHG emissions from soil [93]. Past studies demonstrated elevated activity of organic carbon and nitrogen and the control of ammonium nitrogen when employing a combination of organic and inorganic fertilizers. This also accounted for N_2O emissions being lower when using a microbial organic fertilizer than when using a microbial decomposition treatment because the organic matter in the microbial organic fertilizer improved nitrogen fixation [94]. By altering the functional gene of the denitrifying organisms, microbial fertilizer can also change how soil denitrification occurs. This could limit the N_2O flux when a microbial fertilizer is used [95]. Soil compaction can be reduced by using treatments like biochar and bio-digestate. The soil could become softer and more fertile thanks to biochar [96]. Significant changes are made to the bulk density of soil by biochar. Biochar can help to sequester carbon by retaining the nutrients in the soil, which is one of its most advantageous characteristics from an environmental standpoint. Charcoal requires a long time to break down because of its stable, aromatic carbon structure [97]. Comparatively, ordinary compost quickly breaks down into CO_2 and NH_4^+ after being digested by soil microbes. The application of microbial fertilizers preserved soil moisture in addition to soil heat. Microbial fertilizers decreased the N_2O and CH_4 fluxes throughout the corn cropping season and the CO_2 emissions by 6.9–18.9% compared to chemical fertilizers [98,99]. However, under the application of microbial fertilizer, the soil temperature provided a more accurate description of the N_2O and CH_4 fluxes than the soil moisture. The impact of microbial fertilizers on corn output was more substantial when using different tillage techniques. It is important to note that CH_4 is 28 times more effective than CO_2 at blocking heat in the atmosphere. As a result, biochar can help lessen the GHG emissions into the atmosphere, which can help reduce the impact of climate change through a sustainable approach to agricultural practices [100]. Figure 6 illustrates the advantages of biochar application to agricultural soil. Therefore, adding biochar to the soil is the most excellent method to reduce the amount of CH_4 released [101]. According to one investigation, biochar is one of the most efficient ways to lower GHG emissions. A study [102] suggested that bio-digestate, biochar, and the combined application of biochar and an inorganic fertilizer significantly increased the sequestration of the soil's organic carbon. Nitrification inhibitors can significantly reduce the soil CO_2 and N_2O [103]. These can help to reduce the soil GHG emissions without affecting the productivity of crops.

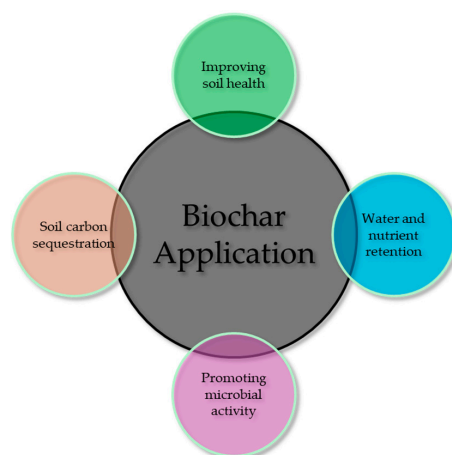


Figure 6. Benefits of biochar application to agricultural soil.

4.3. Cropping System Management

Implementing crop rotation with legumes is a significant strategy for mitigating GHG emissions by decreasing the dependence on nitrogen inputs [104]. However, it is important to note that the nitrogen derived from legumes can also contribute to N₂O emissions [105]. Long-term research in Illinois suggests that compared to the continuous cultivation of corn or soybean, a rotation of corn and soybean can improve the yields, while simultaneously reducing GHG emissions by almost 2 kg N ha⁻¹ yr⁻¹ [106]. The traditional method (i.e., the monoculture), which lacks the intercropping of vegetables, is an intriguing replacement that maximizes the utilization of available land. The system becomes more ecologically balanced because of increased output, optimal use of the inputs, and maximum utilization of environmental resources [107]. It was found that intercropping lettuce, cucumber, and tomato is preferable to practicing monocultures from an economic standpoint [105]. Studies have indicated that the impact of vegetable cultivation on global warming is notable, primarily attributed to substantial greenhouse gas emissions arising from energy consumption, irrigation, agricultural practices, and fertilizer usage, with specific emphasis on greenhouse environments [108,109]. Conservation management practices can improve the productivity of crops and lower the carbon footprint by 31% [110]. Additionally, adopting less intensive farming methods, which entail a reduced pesticide and input usage, can further contribute to emission reduction and lower the overall cost associated with GHG generation [111].

4.4. Tillage Management

Different tillage practices and the soil's physical structure disruption can influence the soil environment and fertility [112]. Furthermore, alterations in soil temperature and moisture resulting from disturbances in the soil physical structure directly affect the GHG emissions from the soil by impacting microbial decomposition and root respiration processes [113]. The impact of tillage decisions on crop yield remains uncertain; however, the extent of yield losses due to no-tillage depends on factors such as soil texture and increased wetness. No-tillage practices were found to increase CO₂ emissions by 7.1%, CH₄ emissions by 20.8%, and N₂O emissions by 12.0% [114]. Reduced tillage techniques like zero tillage offer better aeration and reduce soil moisture, which suggests that they may be a better option than conventional tillage, combining higher levels of soil organic matter and lower N₂O emissions. Reduced tillage has the potential as a best management practice to reduce GHGs because there is evidence that it increases carbon storage while decreasing N₂O emissions and has generally beneficial economic effects [115]. The implementation of deep tillage, despite the intention to conserve soil heat and moisture, proved ineffective, as it led to an increase in overall CO₂ emissions by 4.9–37.7%. However, deep tillage does not have a noticeable effect on total N₂O and CH₄ emissions. Soil CO₂ emissions under deep tillage were significantly greater compared to those under rotational tillage, underscoring the considerable impact of deep tillage on soil CO₂ emissions [116].

4.5. Irrigation and Drainage Management

Drought is recognized as one of the most economically burdensome natural disasters for humanity [103], and its intensity is projected to escalate in the future due to global warming fueled by GHG emissions [117]. Catastrophic drought events impact around 35% of the Earth's total land area, affecting over 120 nations and territories [118]. This situation has prompted researchers to devise machine learning and artificial irrigation technologies; however, the energy required for water pumping can significantly increase GHG emissions [119]. Irrigation accounts for more than 70% of the total global water extraction [98,120]. Mulched drip irrigation emerged as a water-saving technique that generates the least integrated GHG emissions compared to other irrigation methods. Excessive irrigation has the potential to contribute to increased emissions of N₂O through denitrification and of CH₄ from the breakdown of soil organic matter in flooded soil [121]. Over a three-year trial period, sprinkler irrigation demonstrated a 40% reduction in N₂O emissions

and an over 99% reduction in CH₄ emissions compared to flooding [122]. Similarly, proper drainage is vital. Drainage holds the potential to decrease the overall GHG emissions from rice paddy systems in the early season [123]. However, it is crucial to note that any nitrogen lost through drainage may be emitted as N₂O [124].

4.6. Manure Management

Factors like temperature, hydrolysis rate, and degradability of organic matter related to slurry are some of the critical contributors towards CH₄ generation. Ref. [125] stated that composting was identified as a technique with substantial potential to mitigate GHG emissions. Carbon loss in a passive aeration treatment in the forms of CO₂ and CH₄ amounted to 73.8 and 6.3 kg CMg^{−1} of manure, respectively, while the active treatment led to losses of 168.0 and 8.1 kg CMg^{−1} of manure. The passive aeration treatment exhibited lower emissions due to insufficient manure breakdown. Ref. [126] suggested that incorporating lignite into compost has the potential to reduce NH₃ emissions by approximately 60%, although further research is required to understand its impact on other GHG emissions. To address GHG emissions, the first strategy should be optimizing the timing of treatment application throughout the growing season to minimize the soil N₂O emissions. Manure digestion to decrease CH₄ emissions during the storage time is also important [127]. These approaches represent targeted measures to address specific aspects of GHG emissions in manure management systems.

4.7. GWP under Agricultural Management Practices

Sustainable agricultural practices can significantly influence GHG emissions and soil carbon sequestration, impacting the GWP of agricultural systems [128]. Bio-resource management enhances soil organic matter through organic amendments like biochar and bio-digestate, which sequester carbon and improve soil health [129]. Nutrient management addresses the excessive use of fertilizers, which leads to high N₂O emissions [130]. Microbial fertilizers enhance nitrogen fixation and reduce denitrification, further decreasing N₂O emissions. Cropping system management with crop rotation and intercropping reduces the reliance on synthetic fertilizers and optimizes the land use, lowering GHG emissions. Less intensive farming methods also help in reducing emissions and costs [131]. Tillage management impacts soil structure and emissions and can play an important role in improving aeration and soil organic matter while decreasing N₂O emissions [132]. However, deep tillage can significantly increase CO₂ emissions, indicating the need for a careful technique selection based on soil and climate conditions. Efficient irrigation methods like drip and sprinkler systems significantly reduce N₂O and CH₄ emissions compared to traditional flooding. Proper drainage also decreases the GHG emissions from rice paddies by preventing anaerobic conditions that lead to CH₄ production. Manure management controls CH₄ and N₂O emissions through techniques like composting and optimizing the application timing [133]. Sustainable agricultural management techniques are essential for reducing the GWP, enhancing soil carbon sequestration, and lowering the GHG emissions [134]. These practices promote a sustainable and climate-resilient agriculture while improving soil health and productivity and can reduce the GWP.

5. Summary

GHG emissions are intricately linked to fundamental societal necessities, particularly in the realms of food and energy. Unless substantial efforts are undertaken to curb these emissions, they are poised to persistently increase, contributing to global warming and environmental changes. Agriculture is uniquely positioned as both a contributor and a casualty of climate change. Despite an emerging trend to mitigate GHG emissions in agriculture, such measures must not compromise farm production and profitability. This comprehensive study investigated the key drivers influencing GHG emissions from agricultural soils, focusing on proximal and distal factors such as land use, nutrient availability, humidity, temperature, soil pH, and fertilizer and organic amendments. These drivers have

an impact on soil GHG emissions. A standard approach for calculating the GHG budget or estimating soil emission rates is needed. This study described measurement techniques, including chamber systems, eddy covariance methods, and spaceborne approaches.

By pursuing sustainable practices that harmonize ecological objectives with the practical needs of agriculture, it becomes possible to navigate the intricate interplay between societal needs, environmental responsibility, and the agricultural industry's economic viability in the face of a changing climate. On a global scale, comprehensive assessments involve integrating location-specific data, regional data acquisition, reporting, and satellite information through pertinent models. An effective manure management can significantly reduce soil GHG emissions while boosting crop yields when combined with tillage, irrigation, and drainage practices. Bio-resources and nutrient management increase environmental sustainability and reduce emissions from agricultural soils. Creating efficient strategies entails evaluating the GWP and incorporating diverse mitigation techniques. These include managing tillage, irrigation, drainage and overseeing cropping systems, bio-resources, nutrients, and manure. Additionally, the economic viability of these approaches should be carefully assessed.

Despite the promise of mitigating GHG emissions, there exists a disparity between the potential of technologies and the actual reduction achieved, primarily due to barriers in strategy implementation. These barriers include elements like policies related to climate change, as well as educational and economic limitations. To bridge this gap, governments must prioritize climate-smart agriculture technologies, emphasizing the enhancement of input use efficiency within the agricultural sector. By addressing these obstacles and fostering the adoption of environmentally conscious agricultural practices, a more prosperous and impactful approach to GHG mitigation can be realized.

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